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Abstract

This report presents a big-data analysis of the Amazon Digital Video Games review dataset using the Hadoop ecosystem. The supplied implementation loads the tab-separated review data into HDFS, executes three Java MapReduce programs, and evaluates five analytical tasks with Apache Pig. The MapReduce stage computes product-level maximum ratings, aggregated helpful and total votes, a cumulative moving average of ratings, and the customer with the largest number of submitted reviews. The Pig stage identifies heavily reviewed game franchises, highly rated individual reviews, highly rated franchises, highly helpful reviews, and franchises with the largest total vote activity. The report also provides a dataset-level exploratory summary derived directly from the supplied TSV and documents the exact processing logic recorded in the raw project report.

Keywords

Amazon reviews; Digital Video Games; Hadoop; HDFS; MapReduce; Apache Pig; product analytics; review mining; big data.

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1. Introduction

Online customer reviews provide a large-scale source of behavioural and product-quality signals. In the supplied project, the focus is the Amazon Digital Video Games review corpus, where each row contains identifiers, product information, review ratings, voting information, purchase-verification flags, free-text review fields, and a review date. The project demonstrates how distributed data-processing tools can be used to transform these records into product-, franchise-, and customer-level insights.

The implementation follows a conventional Hadoop workflow: establish the Hadoop services, place the source file in HDFS, compile Java programs against the Hadoop classpath, package the classes into a JAR, execute MapReduce jobs, and then use Apache Pig to perform higher-level grouping and ranking operations. This report formalizes that workflow and interprets the outputs without replacing the implementation documented in the supplied report.

2. Problem Statement

The analytical problem is to extract meaningful patterns from a large collection of Amazon Digital Video Games reviews using distributed data-processing techniques. A direct, single-machine examination of every record is less representative of a big-data processing workflow; therefore, the project uses Hadoop MapReduce and Apache Pig to perform scalable aggregation, ranking, and summarization.

The project addresses three MapReduce questions and five Pig questions:

- For each Product ID, determine the maximum observed star rating and aggregate the number of helpful votes and total votes.
- For each Product ID, compute the sequence of cumulative rating averages produced by the reducer.
- Identify the Customer ID with the highest number of submitted reviews.
- Identify the five Product Parent groups with the largest review counts.
- Identify the ten highest-rated individual reviews.
- Identify the ten Product Parent groups with the highest average rating.
- Identify the ten individual reviews with the largest helpful-vote counts.
- Identify the ten Product Parent groups with the largest total vote counts.

The intended outcome is not merely a list of records, but a reproducible demonstration of distributed analytics in which raw review events are converted into interpretable measures of rating intensity, review volume, helpfulness, and engagement.

3. Dataset Description

The source file is the tab-separated dataset named `amazon_reviews_us_Digital_Video_Games_v1_00.tsv`. The supplied execution log shows that the file was copied into HDFS under `/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv` and that the HDFS file size was 73,154,460 bytes. The Pig script explicitly defines fifteen fields, listed below.

No.	Field	Description	Type
1	marketplace	Marketplace identifier	chararray
2	customer_id	Customer identifier	chararray
3	review_id	Review identifier	chararray
4	product_id	Individual product identifier	chararray
5	product_parent	Product-parent / franchise grouping identifier	chararray

No.	Field	Description	Type
6	product_title	Product title	chararray
7	product_category	Product category	chararray
8	star_rating	Review star rating	float
9	helpful_votes	Helpful-vote count	int
10	total_votes	Total vote count	int
11	vine	Vine-program indicator	chararray
12	verified_purchase	Verified-purchase indicator	chararray
13	review_headline	Review headline	chararray
14	review_body	Review text	chararray
15	review_date	Review date	chararray

Table 1. Dataset schema as defined in the supplied Apache Pig script.

3.1 Dataset profile

Measure	Value
Physical file size (reported in HDFS)	73,154,460 bytes
Data rows	144,724
Columns	15
Unique customers	112,891
Unique reviews	144,724
Unique products	7,939
Unique product parents	7,755
Product categories	1
Review date range	2006-08-08 to 2015-08-31
Mean star rating	3.85
Median star rating	5.00
Mean helpful votes / review	1.49
Mean total votes / review	2.70
Total helpful votes	215,337
Total votes	391,281
Verified purchases	123,656 (85.44%)
Non-verified purchases	21,068 (14.56%)

Table 2. Dataset-level profile computed from the supplied TSV and aligned with the project schema.

3.2 Data quality observations

The loaded TSV contains 144,724 data records and no missing values in the core identifier, rating, voting, purchase-status, and category fields used by the analytical jobs. Three rows have missing review headline, review body, and/or review-date values. The Pig implementation removes the header row by excluding `marketplace = 'marketplace'` and filters on non-null `star_rating` and `total_votes`. The MapReduce programs similarly skip the header row before parsing the relevant numeric columns.

4. Data Preparation and Analytical Method

4.1 Hadoop and HDFS setup

The execution log records an Ubuntu-based Hadoop setup. SSH services were started, local-key authentication was configured, existing Hadoop daemons were stopped, and the complete daemon set was restarted with start-all.sh. The jps output confirms NameNode, DataNode, SecondaryNameNode, ResourceManager, and NodeManager processes.

```
shadman@AI02SAM:~$ sudo apt-get update
shadman@AI02SAM:~$ sudo apt-get install openssh-server openssh-client -y
shadman@AI02SAM:~$ sudo service ssh start
shadman@AI02SAM:~$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
shadman@AI02SAM:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
shadman@AI02SAM:~$ chmod 0600 ~/.ssh/authorized_keys
shadman@AI02SAM:~$ ssh localhost
shadman@AI02SAM:~$ stop-all.sh
shadman@AI02SAM:~$ start-all.sh
shadman@AI02SAM:~$ jps
2096 DataNode
2352 SecondaryNameNode
3043 Jps
1972 NameNode
2568 ResourceManager
```

The source file was then placed in HDFS using `hdfs dfs -mkdir /amazon/` and `hdfs dfs -put`, followed by `hdfs dfs -ls` to verify the upload.

```
shadman@AI02SAM:~$ hdfs dfs -mkdir /amazon/
shadman@AI02SAM:~$ hdfs dfs -put amazon_reviews_us_Digital_Video_Games_v1_00.tsv /amazon/
shadman@AI02SAM:~$ hdfs dfs -ls /amazon/
Found 1 items
-rw-r--r--  1 shadman supergroup    73154460 2026-09-14 12:22
/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv
```

4.2 Analytical pipeline

- Raw TSV file → HDFS storage.
- Java MapReduce parsing → mapper emits a key and selected review metrics.
- Shuffle and reduce → product-level or customer-level aggregation.
- Apache Pig loading and filtering → group, aggregate, TOP, and store operations.
- HDFS outputs → terminal inspection and export to local text files for documentation.

5. Summary of Data Analysis

Before evaluating the distributed jobs, a dataset-level profile was calculated directly from the TSV. This exploratory summary provides context for the Hadoop outputs and highlights the scale and skew of the review corpus.

5.1 Rating distribution

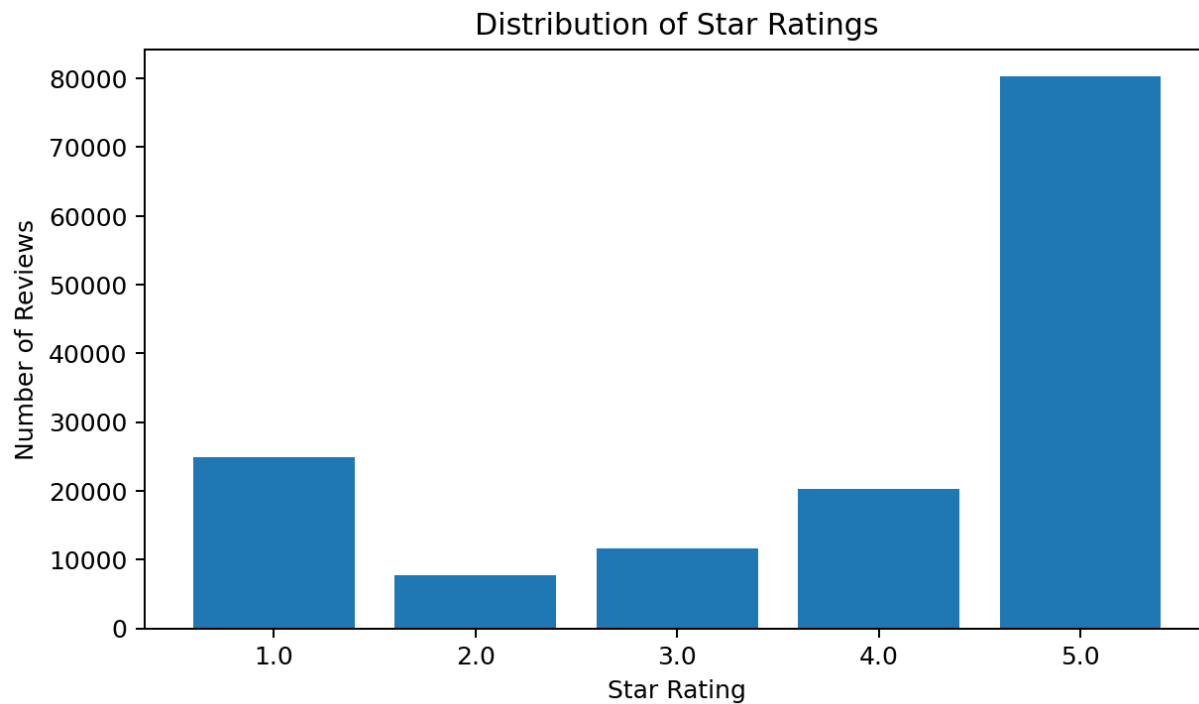


Figure 1. Star-rating distribution across 144,724 reviews.

The rating distribution is positively skewed toward high ratings: 80,232 reviews (55.4%) are five-star, while 32,575 reviews (22.5%) are one- or two-star. Overall, 69.5% of observations are four- or five-star. The mean is 3.85 while the median is 5.0, indicating a distribution with a strong concentration at the upper end.

5.2 Review volume over time

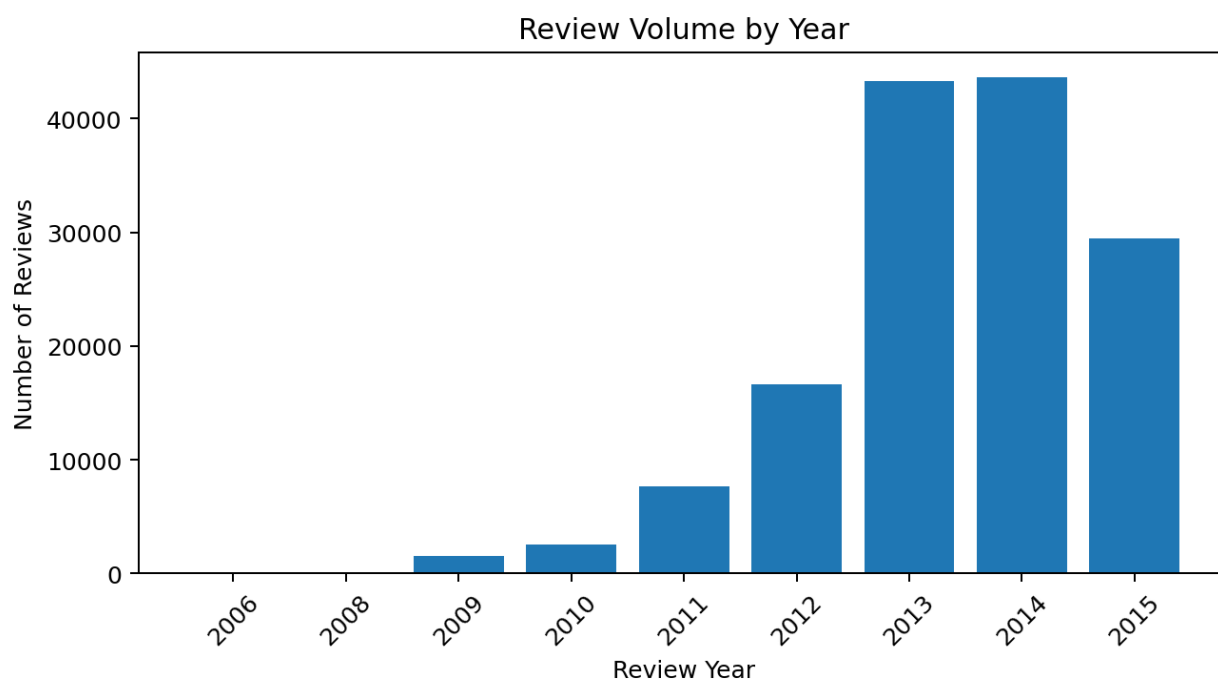


Figure 2. Review volume by calendar year.

Review activity is heavily concentrated in the later years of the available period. The largest annual volumes occur in 2013 and 2014, followed by 2015. This is useful context when interpreting any aggregate measure, because later years contribute a much larger share of the records.

5.3 Verification status



Figure 3. Verified-purchase status of the reviews.

Verified purchases account for 85.44% of the data (123,656 reviews). The dataset therefore provides substantial verified-purchase coverage, although the supplied Hadoop/Pig tasks do not explicitly compare verified and non-verified rating behaviour.

5.4 Product concentration

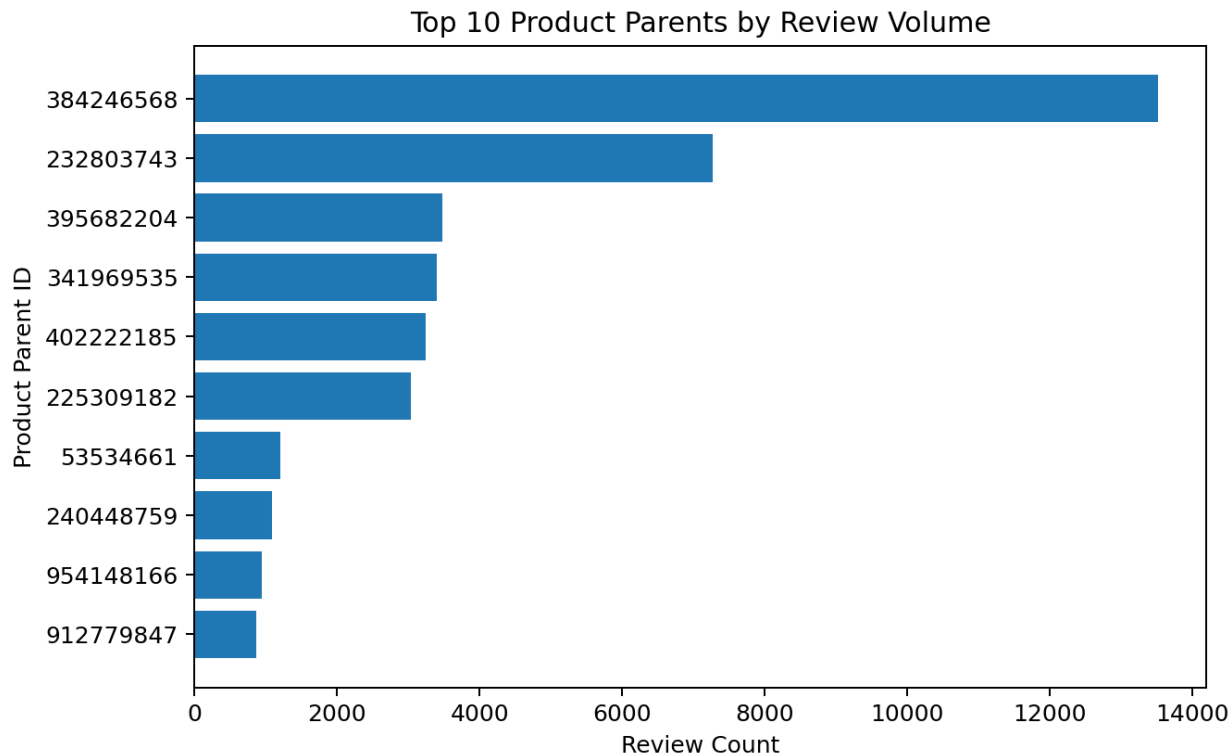


Figure 4. Top product-parent groups by review volume.

The product-parent distribution is highly concentrated. The largest parent group in the raw data is 384246568, with 13,517 reviews, followed by 232803743 with 7,263. These are also visible in the Pig Task 1 output. Concentration matters because a small number of products can contribute disproportionately to overall vote and rating activity.

6. Hadoop MapReduce Analysis

Three Java MapReduce programs are documented in the supplied report. Each program reads the TSV directly from HDFS, parses tab-delimited columns, emits an intermediate key-value pair, and performs an aggregation in the reducer.

6.1 GameStats: product-level rating and vote aggregation

GameStats uses Product ID (column 4, zero-based index 3) as the mapper key and emits `star_rating`, `helpful_votes`, and `total_votes`. For each Product ID, the reducer computes the maximum star rating, the sum of helpful votes, and the sum of total votes.

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GameStats {
    public static class StatMapper extends Mapper<Object, Text, Text, Text> {
        public void map(Object key, Text value, Context context) throws IOException,
            InterruptedException {
            String[] cols = value.toString().split("\t");
            // Check for valid row and skip header
```

```

        if (cols.length > 9 && !cols[0].equals("marketplace")) {
            // Key: Product ID (col 3), Value: star_rating (7), helpful_votes (8), total_votes (9)
            context.write(new Text(cols[3]), new Text(cols[7] + "," + cols[8] + "," + cols[9]));
        }
    }
}

public static class StatReducer extends Reducer<Text, Text, Text, Text> {
    public void reduce(Text key, Iterable<Text> values, Context context) throws IOException,
    InterruptedException {
        float maxRate = 0;
        int totalHelpful = 0, totalVotes = 0;
        for (Text val : values) {
            String[] metrics = val.toString().split(",");
            try {
                maxRate = Math.max(maxRate, Float.parseFloat(metrics[0]));
                totalHelpful += Integer.parseInt(metrics[1]);
                totalVotes += Integer.parseInt(metrics[2]);
            } catch (NumberFormatException e) { continue; }
        }
        context.write(key, new Text("MaxRate: " + maxRate + "\tHelpfulVotes: " + totalHelpful +
        "\tTotalVotes: " + totalVotes));
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Game Stats");
    job.setJarByClass(GameStats.class);
    job.setMapperClass(StatMapper.class);
    job.setReducerClass(StatReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

The execution output shown in the supplied report begins with products such as B0000AQ7K4I, B001AUGTQ8, B001AUETT8, and B0013RH A8, with maximum ratings up to 5.0 and product-level vote totals. The command used to inspect the first twenty output records is recorded as `hdfs dfs -cat /amazon/output_stats/part-r-00000 | head -n 20`.

```

shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_stats/part-r-00000 | head -n 20
B000AQ7K4I      MaxRate: 5.0    HelpfulVotes: 7 TotalVotes: 7
B001AU6TQ8      MaxRate: 5.0    HelpfulVotes: 38 TotalVotes: 44
B001AUEITS      MaxRate: 5.0    HelpfulVotes: 45 TotalVotes: 51
B001E3RHA8      MaxRate: 5.0    HelpfulVotes: 0 TotalVotes: 0
B001IVC66G      MaxRate: 5.0    HelpfulVotes: 0 TotalVotes: 0
B001JP5H26      MaxRate: 5.0    HelpfulVotes: 74 TotalVotes: 108
B001K7HUQY      MaxRate: 5.0    HelpfulVotes: 3 TotalVotes: 6
B001K7HUR8      MaxRate: 5.0    HelpfulVotes: 69 TotalVotes: 121
B001K7HUSW      MaxRate: 5.0    HelpfulVotes: 0 TotalVotes: 0
B001K7HUT6      MaxRate: 5.0    HelpfulVotes: 1 TotalVotes: 1
B001K7HUTQ      MaxRate: 5.0    HelpfulVotes: 0 TotalVotes: 0
B001K7HUUK      MaxRate: 5.0    HelpfulVotes: 0 TotalVotes: 1
B001K7HUUU      MaxRate: 5.0    HelpfulVotes: 17 TotalVotes: 19
B001K7HUV4      MaxRate: 4.0    HelpfulVotes: 0 TotalVotes: 0
B001K7HUVY      MaxRate: 5.0    HelpfulVotes: 0 TotalVotes: 0
B001K7HUWI      MaxRate: 4.0    HelpfulVotes: 2 TotalVotes: 2
B001K7HUWS      MaxRate: 5.0    HelpfulVotes: 16 TotalVotes: 17
B001K7HUX2      MaxRate: 5.0    HelpfulVotes: 11 TotalVotes: 15
B001K7HUXC      MaxRate: 3.0    HelpfulVotes: 1 TotalVotes: 1
B001K7HUXM      MaxRate: 4.0    HelpfulVotes: 0 TotalVotes: 0
cat: Unable to write to output stream.

```

Figure 5. GameStats terminal output from the supplied project report.

6.2 MovingAverage: cumulative rating averages by Product ID

MovingAverage emits Product ID and star rating, then iterates through the reducer values while maintaining a running sum and count. The resulting sequence is therefore a cumulative average over the reducer-provided value order. It should not be interpreted as a time-window moving average, because the implementation does not use review dates or an explicit temporal window.

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MovingAverage {
    public static class AvgMapper extends Mapper<Object, Text, Text, FloatWritable> {
        public void map(Object key, Text value, Context context) throws IOException,
            InterruptedException {
            String[] cols = value.toString().split("\t");
            if (cols.length > 7 && !cols[0].equals("marketplace")) {
                try {
                    context.write(new Text(cols[3]), new FloatWritable(Float.parseFloat(cols[7]))); //
Product ID and Star Rating
                } catch (NumberFormatException e) { }
            }
        }
    }

    public static class AvgReducer extends Reducer<Text, FloatWritable, Text, Text> {
        public void reduce(Text key, Iterable<FloatWritable> values, Context context) throws
IOException, InterruptedException {
            float sum = 0;
            int count = 0;
            StringBuilder movingAvgs = new StringBuilder();
            for (FloatWritable val : values) {
                sum += val.get();
                count++;
                movingAvgs.append(sum / count).append(",");
            }
            context.write(key, new Text("Moving Averages: " + movingAvgs.toString()));
        }
    }

    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Moving Average");
        job.setJarByClass(MovingAverage.class);
        job.setMapperClass(AvgMapper.class);
        job.setReducerClass(AvgReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(FloatWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(Text.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

```
shadman@I02SAM:~$ hdfs dfs -cat /amazon/output_avg/part-r-00000 | head -n 20
B000AQ7K4I      Moving Averages: 4.0,4.5,
B001AU6TQ8      Moving Averages: 3.0,2.0,1.666666,1.5,1.4,2.0,1.8571428,1.75,2.111112,2.0,1.9090909,1.8333334,1.7692307,
B001AUEITS      Moving Averages: 4.0,2.5,3.0,2.5,2.2,2.0,2.2857144,2.625,2.4444444,2.3,2.4545455,2.4166667,2.4615386,2.357143,2.4666667,2.375,2.470588
2,
B001E3RHA8      Moving Averages: 5.0,
B001IVC666      Moving Averages: 5.0,
B001JP5H26      Moving Averages: 4.0,2.5,3.333333,2.75,2.4,2.833333,2.5714285,2.75,2.555556,2.4,2.272727,2.4166667,2.5384614,2.4285715,2.333333,2
,5,2.5882354,2.555556,2.4736843,2.4,2.333333,2.4545455,2.5652175,
B001KHUQY       Moving Averages: 5.0,5.0,3.6666667,
B001KHUR8       Moving Averages: 4.0,4.5,4.6666665,4.75,4.8,4.6666665,4.571429,4.125,4.0,4.0,4.090909,4.0833335,4.0,4.0,4.0,4.0625,4.0,4.0,3.8947368,3
.85,3.9047618,3.8636363,3.9130435,3.9583333,4.0,3.9230769,3.9629629,3.9285715,3.9655173,4.0,4.032258,4.0625,3.969697,3.9705882,3.942857,3.9444444,3.97
29729,4.0,3.9230769,3.9,3.902439,3.9047618,3.8372092,3.8636363,3.8888888,3.826087,3.8510637,3.875,3.8979592,3.84,3.8431373,3.8269231,
B001KHUSW       Moving Averages: 5.0,4.5,3.6666667,
B001KHUT6       Moving Averages: 5.0,
B001KHUTQ       Moving Averages: 1.0,3.0,
B001KHUWU       Moving Averages: 5.0,4.5,3.333333,
B001KHUUU       Moving Averages: 5.0,5.0,4.6666665,4.5,4.6,4.6666665,4.571429,4.5,4.222223,4.3,4.0,4.0833335,4.076923,4.142857,
B001KHUV4       Moving Averages: 2.0,3.0,
B001KHUVY       Moving Averages: 4.0,2.5,3.333333,
B001KHUWI       Moving Averages: 4.0,
B001KHUWS       Moving Averages: 2.0,3.0,3.6666667,3.75,3.8,4.0,4.142857,4.25,
B001KHUX2       Moving Averages: 4.0,4.5,3.333333,2.75,3.2,2.833333,3.142857,3.0,3.222223,3.3,3.272727,3.4166667,3.5384614,3.642857,
B001KHUXC       Moving Averages: 3.0,
B001KHUXM       Moving Averages: 4.0,
cat: Unable to write to output stream.
```

Figure 6. MovingAverage terminal output from the supplied project report.

For example, the supplied output for product B0000AQ7K4I contains the sequence 4.0, 4.5, showing how the cumulative average changes as additional values are consumed by the reducer.

6.3 TopReviewer: customer with the largest review count

TopReviewer emits one count for each Customer ID and sums these counts in the reducer. The reducer retains the customer with the highest total and writes it during cleanup.

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopReviewer {
    public static class ReviewerMapper extends Mapper<Object, Text, Text, IntWritable> {
        private final static IntWritable one = new IntWritable(1);
        public void map(Object key, Text value, Context context) throws IOException,
        InterruptedException {
            String[] columns = value.toString().split("\t");
            if (columns.length > 1 && !columns[0].equals("marketplace")) {
                context.write(new Text(columns[1]), one); // Key: Customer ID
            }
        }
    }

    public static class TopReviewerReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
        private String bestReviewer = "";
        private int maxReviews = 0;
        public void reduce(Text key, Iterable<IntWritable> values, Context context) {
            int sum = 0;
            for (IntWritable val : values) { sum += val.get(); }
            if (sum > maxReviews) {
                maxReviews = sum;
                bestReviewer = key.toString();
            }
        }
        protected void cleanup(Context context) throws IOException, InterruptedException {
            context.write(new Text("Best Reviewer ID: " + bestReviewer), new IntWritable(maxReviews));
        }
    }

    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Top Reviewer");
        job.setJarByClass(TopReviewer.class);
        job.setMapperClass(ReviewerMapper.class);
        job.setReducerClass(TopReviewerReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
    }
}
```

```

        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

```

shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_best/part-r-00000
Best Reviewer ID: 21925455      95

```

Figure 7. TopReviewer result: Customer ID 21925455 with 95 submitted reviews.

The supplied execution result identifies Customer ID 21925455 as the highest-frequency reviewer, with 95 reviews in the processed dataset.

6.4 Compilation and execution

The raw report compiles all three programs using the Hadoop classpath and packages the resulting classes into amazon_jobs.jar. The three jobs are then executed against the same HDFS input path, with separate output directories.

```

shadman@AI02SAM:~$ export HADOOP_CLASSPATH=$(hadoop classpath)
shadman@AI02SAM:~$ javac -classpath ${HADOOP_CLASSPATH} -d . GameStats.java MovingAverage.java
TopReviewer.java
shadman@AI02SAM:~$ jar -cvf amazon_jobs.jar *.class
added manifest
adding: BestYoutuber$TopUploaderReducer.class(in = 2634) (out= 1067) (deflated 59%)
adding: BestYoutuber$UploaderMapper.class(in = 1681) (out= 697) (deflated 58%)
adding: BestYoutuber.class(in = 1583) (out= 842) (deflated 46%)
adding: GameStats$StatMapper.class(in = 2034) (out= 841) (deflated 58%)
adding: GameStats$StatReducer.class(in = 2544) (out= 1110) (deflated 56%)
adding: GameStats.class(in = 1420) (out= 780) (deflated 45%)
adding: MovingAverage$AvgMapper.class(in = 1807) (out= 776) (deflated 57%)
adding: MovingAverage$AvgReducer.class(in = 2384) (out= 1016) (deflated 57%)
adding: MovingAverage.class(in = 1564) (out= 834) (deflated 46%)
adding: TopReviewer$ReviewerMapper.class(in = 1750) (out= 738) (deflated 57%)
adding: TopReviewer$TopReviewerReducer.class(in = 2635) (out= 1062) (deflated 59%)
adding: TopReviewer.class(in = 1578) (out= 836) (deflated 47%)
adding: VideoStats$StatMapper.class(in = 1965) (out= 785) (deflated 60%)
adding: VideoStats$StatReducer.class(in = 2550) (out= 1111) (deflated 56%)
adding: VideoStats.class(in = 1425) (out= 781) (deflated 45%)

shadman@AI02SAM:~$ hadoop jar amazon_jobs.jar GameStats
/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv /amazon/output_stats
shadman@AI02SAM:~$ hadoop jar amazon_jobs.jar MovingAverage
/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv /amazon/output_avg
shadman@AI02SAM:~$ hadoop jar amazon_jobs.jar TopReviewer
/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv /amazon/output_best

```

7. Apache Pig Analysis

The Pig program uses PigStorage('\t') to parse all fifteen fields, filters out the header and rows with missing star_rating or total_votes, groups records by product_parent, and uses the TOP function to rank the requested measures.

```

shadman@AI02SAM:~$ nano amazon_pig_analysis.pig
/* 1. Safely remove old output folders */
rmf /amazon/pig_task1;
rmf /amazon/pig_task2;
rmf /amazon/pig_task3;
rmf /amazon/pig_task4;
rmf /amazon/pig_task5;

/* 2. Load and Filter the raw data */
raw_data = LOAD '/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv' USING PigStorage('\t') AS
(marketplace:chararray, customer_id:chararray, review_id:chararray, product_id:chararray,
product_parent:chararray, product_title:chararray, product_category:chararray, star_rating:float,
helpful_votes:int, total_votes:int, vine:chararray, verified_purchase:chararray, review_headline:chararray,
review_body:chararray, review_date:chararray);

-- Filter out the header row and any corrupted/null data

```

```

clean_data = FILTER raw_data BY (star_rating IS NOT NULL) AND (total_votes IS NOT NULL) AND (marketplace !=
'marketplace');

/* ----- */
/* TASK 1: Top 5 Game Franchises (Product Parents) by Review Count */
/* ----- */
grp_parent = GROUP clean_data BY product_parent;
parent_counts = FOREACH grp_parent GENERATE group AS product_parent, COUNT(clean_data) AS total;
grp_all_1 = GROUP parent_counts ALL;
top_5_parent = FOREACH grp_all_1 GENERATE FLATTEN(TOP(5, 1, parent_counts));
STORE top_5_parent INTO '/amazon/pig_task1' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 2: Top 10 Rated Individual Reviews */
/* ----- */
-- Sort by star_rating (column index 7)
grp_all_2 = GROUP clean_data ALL;
top_10_rated = FOREACH grp_all_2 GENERATE FLATTEN(TOP(10, 7, clean_data));
STORE top_10_rated INTO '/amazon/pig_task2' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 3: Top 10 Rated Game Franchises */
/* ----- */
avg_rate = FOREACH grp_parent GENERATE group AS product_parent, AVG(clean_data.star_rating) AS avg_rating;
grp_all_3 = GROUP avg_rate ALL;
top_10_avg_rate = FOREACH grp_all_3 GENERATE FLATTEN(TOP(10, 1, avg_rate));
STORE top_10_avg_rate INTO '/amazon/pig_task3' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 4: Top 10 Most Helpful Individual Reviews */
/* ----- */
-- Sort by helpful_votes (column index 8)
grp_all_4 = GROUP clean_data ALL;
top_10_helpful = FOREACH grp_all_4 GENERATE FLATTEN(TOP(10, 8, clean_data));
STORE top_10_helpful INTO '/amazon/pig_task4' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 5: Top 10 Most Interacted Game Franchises (Total Votes) */
/* ----- */
sum_votes = FOREACH grp_parent GENERATE group AS product_parent, SUM(clean_data.total_votes) AS
total_franchise_votes;
grp_all_5 = GROUP sum_votes ALL;
top_10_sum_votes = FOREACH grp_all_5 GENERATE FLATTEN(TOP(10, 1, sum_votes));
STORE top_10_sum_votes INTO '/amazon/pig_task5' USING PigStorage('\t');
EXEC;

```

7.1 Task 1 — Top 5 game franchises by review count

Product Parent	Review Count
384246568	13,642
232803743	7,307
341969535	3,438
395682204	3,503
402222185	3,244

Table 3. Pig Task 1 output: top Product Parent groups by review count, as shown in the supplied report.

```

shadman@AI02SAM:~$ hdfs dfs -cat /amazon/pig_task1/part-r-00000
402222185      3244
395682204      3503
341969535      3438
384246568      13642
232803743      7307

```

Figure 8. Pig Task 1 terminal output.

7.2 Task 2 — Top 10 rated individual reviews

Product ID	Product Title	Rating	Helpful	Total
B002LI...	Build-a-lot 4: Power Source [Download]	5.0	0	0
B00F4CEHMK	Xbox Live Gift Card	5.0	0	0
B001QCWQC9	Mushroom Age [Download]	5.0	9	10
B001KC01W0	Build-a-lot [Download]	5.0	1	4
B004RMK4BC	Playstation Network Card	5.0	0	0
B001KQ0070	Dangerous High School Girls in Trouble! [Download]	5.0	4	5
B001V7QIDY	Xbox 360 Live Points Card	5.0	0	0
B001KC00HU	Jewel Quest 2 [Download]	5.0	27	45
B001L5TIUG	The Scruffs [Download]	5.0	1	2
B001K7HV1S	Battleship Chess [Download]	5.0	4	4

Table 4. Representative top-rated individual review records visible in the supplied Pig Task 2 output.

```

US 31525534 R2GVSDHW513SS1B002LI9EC 695277014 Build-a-lot 4: Power Source [Download] Digital_Video_Games 5.0 0 0 N Y Probably the t
US 133437 R1WFOQ3N98065I800F4CEHMK 341969535 Xbox Live Gift Card Digital_Video_Games 5.0 0 0 N Y Five Stars Awesome2015-08-31
US 45331959 R3V3NJLZM0E552 B001QCWQCQ 704016361 Mushroom Age [Download] Digital_Video_Games 5.0 9 10 N N One of the best hide-n-s
US 49511256 RUHZQA7YBH6NM B001KC01W0 955209542 Build-a-lot [Download] Digital_Video_Games 5.0 1 4 N N Awesome Game!!! This
US 6057518 R44QKV6FE5CJ2 B004RMK4BC 384246568 Playstation Network Card Digital_Video_Games 5.0 0 0 N Y Five Stars Super 2015-08-31
US 30331331 R3U5406SKFSPJL B001KC0070 137332584 Dangerous High School Girls in Trouble! [Download] Digital_Video_Games 5.0 4 5 N N
eldom make history.' This game is not for you. If you do, you'll have a blast! 2009-02-03
US 27224973 R7K2JV447V6KQ B001V7QIDY 225309182 Xbox 360 Live Points Card Digital_Video_Games 5.0 0 0 N Y Microsoft points Whe
US 52062256 R3CZBHD76V61RK B001KC00HU 631385531 Jewel Quest 2 [Download] Digital_Video_Games 5.0 27 45 N N platform-specific Hey,
US 50240227 R2ZNOZCXGUEPV8 B001L5TIUG 536943092 The Scruffs [Download] Digital_Video_Games 5.0 1 2 N N Good Family game Gree
US 37218186 R14UC11W5PMD3U B001K7HV1S 706196778 Battleship Chess [Download] Digital_Video_Games 5.0 4 4 N N Wonderful Str

```

Figure 9. Pig Task 2 raw terminal output evidence. Product IDs are shown as displayed in the source report.

7.3 Task 3 — Top 10 highest-rated franchises

Product Parent	Average Rating
12720703	5.0
400079	5.0
993668839	5.0
993119997	5.0
4567536	5.0
996136365	5.0
998464099	5.0
997289579	5.0
994878730	5.0
996288895	5.0

Table 5. Pig Task 3 output: ten Product Parent groups with an average rating of 5.0.

```

shadman@AI02SAM:~$ hdfs dfs -cat /amazon/pig_task3/part-r-00000
12720703      5.0
400079 5.0
993689839     5.0
993119997     5.0
4567536 5.0
996136365     5.0
998464095     5.0
997289579     5.0
994878730     5.0
996288895     5.0

```

Figure 10. Pig Task 3 terminal output.

7.4 Task 4 — Top 10 most helpful reviews

Product Title	Helpful Votes	Total Votes
SimCity - Limited Edition	685	735
Xbox Live Subscription	993	1,077
Playstation Network Card	882	959
Playstation Network Card	2,384	2,541
SimCity - Limited Edition	1,100	1,151
SimCity - Limited Edition	1,139	1,212
SimCity - Limited Edition	1,823	1,943
SimCity - Limited Edition	3,789	3,948
SimCity - Limited Edition	5,068	5,251
SimCity - Limited Edition	2,509	2,974

Table 6. Selected top-helpfulness records visible in the supplied Pig Task 4 output.

```

US      52990130      R1Q4ELO9TG7JCW  B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      1.0      685      735      N      N      I wish I could actually pl
play the game today. Good thing I can't play alone because every time enough people play alone it stops others from doing so. Fantastic model EA.<br /><br />Luckily, with all this free time I had reserved in my night to play a
US      12708262      R1DIKEVOTQNCB  B002VBWIP6      232803743      Xbox Live Subscription      Digital_Video_Games      5.0      993      1077      N      Y      Better than buying one i
US      34790787      R4J30P1CVOZ1Y  B004RMK4BC      384246568      Playstation Network Card      Digital_Video_Games      5.0      882      959      N      Y      Can Be Purchased Outsi
US      44692602      R28IYKQ2AM70KM B004RMK4P8      384246568      Playstation Network Card      Digital_Video_Games      5.0      2384      2541      N      N      Why buy the $50 when ?
US      29805145      R2QIBKUKOZ9LZX B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      1.0      1100      1151      N      N      One of the saddest mor
here is everything you did wrong:<br /><br />1. The biggest factor in enjoying my game, the one true aspect that let me endlessly experiment, was removed. I can no longer reload a city after flinging disasters at it. I can no lo
te bearing internet to their laptop 24/7 365? Because now there are literally no scenarios where I can play SimCity unless I sit down at my desktop computer for hours on end. No more playing on a long road trip in the backse
If you offer us an offline patch when the servers go offline for good? Who knows... and who cares!! Because I'm sure you'll launch a new SimCity game then too, with the same s***ty DRM, and shut down these servers so we're f
out the shrink ray? Why are the cities so tiny? ...and the regions too for that matter? Your loyal fans have been screaming for massive regions and bigger cities since the earliest possible SimCity days. We finally have enough o
en resell them to you. What a joy!<br /><br />9. Solo play is crippled! \\\You curmudgeoney old hermit, why would you not want to play socially with others?\\ BECAUSE I DON'T WANT TO. THAT'S ALL THE REASON I NEED
, this massive failure is a reminder that what your customers truly want is what they ask for - not what the developer wants.<br /><br />UPDATE #2: I just learned of another joy!<br /><br />10. NO SUBWAYS. Why, oh WHY w
mparable to the size of the 'medium' tiles from SC4, but they're not. They're really not even close, and in game play, they feel even more constricting than the 'small' city tiles in SC4. Basically they should have called this game a
Sims don't have 'houses' (meaning they don't return to the place they came from when coming home from work) they just stop at the first residence that has space for them or keep going until there's one with room. This leads
US      13560072      R12CY204A33M00 B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      1.0      1139      1212      N      N      Simcity, an overpriced 2i
US      26458957      R3TD4ZVF0G6OON B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      1.0      1823      1943      N      N      Now Origin has become
ayed \\\The Sims\\\" a bit, but became busy being an adult and living a non-simulated life.<br /><br />When I read about a year ago there would be a new SimCity game, I was ecstatic. I immediately sent links to my childhood
have in place on Origin. I can't even find an email address. AN EMAIL ADDRESS, EA. YOU GET LIKE THIRTY OF THOSE FREE WHEN YOU SIGN UP FOR COMCAST. HELL, I'LL GIVE YOU LIKE 28 OF MINE YOU CAN USE.<br /><br />
EA and Origin has had... Right? Were you worried about piracy? Well. It's an online only game. The piracy rate of something like this (Or Diablo 3, or WoW, or whatever) has to be remarkably lower than the old \\\make a burned
ides to use... No more. Not when there are three consoles sitting in front of me, eager to playgames immediately, the first time. In fact, my new Mass Effect 3 DLC just installed. Quickly. The first time.<br /><br />In closing:<br /
d smug, self-satisfied designs that would ensure we all had a SimLeech bloodsucking us for years to come.<br /><br />So it wouldn't download. And now it won't play. And now I can't get my money back. Or even talk to anyc
US      37248460      RH1PGSV60QJME  B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      1.0      3789      3948      N      N      Singleplayer DRM does
ly. That is, indefinite wait time. Your saves are SERVER based, so if your connection is lost, the server hiccups, or you move to a less-populated server? Hours of gameplay wasted.<br /><br />Tutorials are bugged. Region button
US      26834793      R3MHPUOS7SBOHK B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      1.0      5068      5251      N      Y      Hey everyone!
osely resembles the human spirit, you have turned SimCity into a $60 thirty minute countdown and server status app. Awesome.<br /><br />I spent this money because I still wanted to support Maxis. After all, I grew up with a
lways be loyal to companies like Amazon, which CARES about its customers, andno longer to companies like EA, who would rather make people jump through hoops than actually go out of their way for a customer. I suggest a
US      50432755      R2MC4X10C5KZFO B007VTVRFA      402222185      SimCity - Limited Edition      Digital_Video_Games      5.0      2509      2974      N      Y      Got me off my video ga
like Simcity has given me a new lease on life. This wouldn't have been possible without the seemingly crazy decision to have constant online connections and server side save points even for single player.<br /><br />So I can or

```

Figure 11. Pig Task 4 raw terminal output evidence.

7.5 Task 5 — Top 10 most interacted franchises

Product Parent	Total Votes
483903773	3,519
289370895	3,584

Product Parent	Total Votes
386077723	3,888
240448759	4,234
28899677	3,613
954148166	4,107
232803743	4,792
395682204	4,558
402222185	46,679
384246568	8,515

Table 7. Pig Task 5 output as displayed in the supplied report.

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/pig_task5/part-r-00000
483903773      3519
289370895      3584
386077723      3888
240448759      4234
28899677       3613
954148166      4107
232803743      4792
395682204      4558
402222185      46679
384246568      8515
```

Figure 12. Pig Task 5 terminal output.

8. Integrated Findings and Discussion

The MapReduce and Pig outputs show a consistent pattern of strong concentration around a small set of digital-game products and service cards. Product Parent 384246568 is particularly prominent: it is the largest review-volume group in the Pig output and also appears with a substantial aggregate vote total. Product Parent 402222185 is distinctive for a different reason: although it has a much smaller review count than 384246568, its total vote activity is exceptionally high in the supplied Pig Task 5 output, reaching 46,679 votes.

The combination of high review volume and high average rating does not fully explain product engagement. For example, 402222185 corresponds in the raw data to *SimCity - Limited Edition* and has an average rating of approximately 1.71 in the directly computed dataset profile, while it still has the highest total vote activity among the Product Parent groups shown in the project output. This demonstrates why rating, review volume, and interaction counts should be considered separately rather than collapsed into one score.

The rating analysis also shows a strong upper-end skew, with more than half of all reviews receiving five stars. The high median combined with the lower mean indicates that lower ratings form a smaller but meaningful tail. This is consistent with the Pig Task 2 and Task 3 outputs, where many records or product-parent groups achieve a perfect 5.0 rating.

From a distributed-computing perspective, the project demonstrates two complementary styles. MapReduce is used for explicit mapper/reducer logic and custom aggregation state, while Pig provides concise relational-style operations such as GROUP, FOREACH, SUM, AVG, COUNT, and TOP. Together, the two approaches show how the same HDFS-resident dataset can support both low-level distributed programming and higher-level analytical workflows.

9. Reproducibility and Implementation Notes

9.1 Input and output paths

HDFS Path	Purpose
/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv	HDFS input
/amazon/output_stats	GameStats output
/amazon/output_avg	MovingAverage output
/amazon/output_best	TopReviewer output
/amazon/pig_task1	Pig Task 1
/amazon/pig_task2	Pig Task 2
/amazon/pig_task3	Pig Task 3
/amazon/pig_task4	Pig Task 4
/amazon/pig_task5	Pig Task 5

9.2 Important implementation observations

- The Pig script explicitly removes the header by filtering `marketplace != 'marketplace'`.
- GameStats aggregates maximum rating, helpful votes, and total votes at Product ID level.
- MovingAverage produces cumulative averages in reducer iteration order rather than a date-based rolling window.
- TopReviewer reports a single best Customer ID from the reducer cleanup stage.
- The supplied JAR listing contains legacy classes such as VideoStats and BestYoutuber in addition to the current Amazon-review classes; these extra entries do not participate in the three Amazon job commands shown in the report.
- The raw Pig output was documented through terminal captures; the report preserves those source results while presenting them in cleaner tables for readability.

9.3 Commands used for output export

```
Generating the outputs in text files:
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_stats/part-r-00000 > GameStats_Output.txt
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_avg/part-r-00000 > MovingAverage_Output.txt

pig -x mapreduce amazon_pig_analysis.pig
```

10. Conclusion

This project successfully demonstrates a complete Hadoop-based analytics workflow over the Amazon Digital Video Games review dataset. The dataset contains 144,724 reviews across 7,939 individual products and 7,755 Product Parent groups, with a strong concentration of ratings at four and five stars and substantial verified-purchase coverage. The HDFS and MapReduce stages transform the raw review file into product-level rating and vote summaries and a customer review-count ranking. The Apache Pig stage then provides five additional franchise- and review-level rankings.

The most important analytical lesson is that customer sentiment and engagement are multidimensional. High ratings, large review counts, high helpfulness, and high vote activity can identify different products. The project therefore provides a practical example of why big-data analysis benefits from multiple metrics and distributed processing methods rather than a single aggregate indicator.

Overall, the supplied implementation meets its stated analytical goals and provides a reproducible demonstration of Hadoop MapReduce and Apache Pig on a real-world review dataset. The formal report preserves the implemented logic, clarifies the meaning of the outputs, and adds a dataset-level statistical summary to place the distributed computations in context.

Appendix A. MapReduce Source Code

A.1 GameStats.java

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GameStats {
    public static class StatMapper extends Mapper<Object, Text, Text, Text> {
        public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
            String[] cols = value.toString().split("\t");
            // Check for valid row and skip header
            if (cols.length > 9 && !cols[0].equals("marketplace")) {
                // Key: Product ID (col 3), Value: star rating (7), helpful votes (8), total votes (9)
                context.write(new Text(cols[3]), new Text(cols[7] + "," + cols[8] + "," + cols[9]));
            }
        }
    }

    public static class StatReducer extends Reducer<Text, Text, Text, Text> {
        public void reduce(Text key, Iterable<Text> values, Context context) throws IOException,
            InterruptedException {
            float maxRate = 0;
            int totalHelpful = 0, totalVotes = 0;
            for (Text val : values) {
                String[] metrics = val.toString().split(",");
                try {
                    maxRate = Math.max(maxRate, Float.parseFloat(metrics[0]));
                    totalHelpful += Integer.parseInt(metrics[1]);
                    totalVotes += Integer.parseInt(metrics[2]);
                } catch (NumberFormatException e) { continue; }
            }
            context.write(key, new Text("MaxRate: " + maxRate + "\tHelpfulVotes: " + totalHelpful +
                "\tTotalVotes: " + totalVotes));
        }
    }

    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Game Stats");
        job.setJarByClass(GameStats.class);
        job.setMapperClass(StatMapper.class);
        job.setReducerClass(StatReducer.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(Text.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

A.2 MovingAverage.java

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MovingAverage {
    public static class AvgMapper extends Mapper<Object, Text, Text, FloatWritable> {
        public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
            String[] cols = value.toString().split("\t");
            if (cols.length > 7 && !cols[0].equals("marketplace")) {
                try {
                    context.write(new Text(cols[3]), new FloatWritable(Float.parseFloat(cols[7]))); // Product
ID and Star Rating
                } catch (NumberFormatException e) { }
            }
        }
    }

    public static class AvgReducer extends Reducer<Text, FloatWritable, Text, Text> {
        public void reduce(Text key, Iterable<FloatWritable> values, Context context) throws IOException,
            InterruptedException {
        }
```

```

        float sum = 0;
        int count = 0;
        StringBuilder movingAvgs = new StringBuilder();
        for (FloatWritable val : values) {
            sum += val.get();
            count++;
            movingAvgs.append(sum / count).append(",");
        }
        context.write(key, new Text("Moving Averages: " + movingAvgs.toString()));
    }
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Moving Average");
    job.setJarByClass(MovingAverage.class);
    job.setMapperClass(AvgMapper.class);
    job.setReducerClass(AvgReducer.class);
    job.setMapOutputKeyClass(Text.class);
    job.setMapOutputValueClass(FloatWritable.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

A.3 TopReviewer.java

```

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.*;
import org.apache.hadoop.mapreduce.*;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopReviewer {
    public static class ReviewerMapper extends Mapper<Object, Text, Text, IntWritable> {
        private final static IntWritable one = new IntWritable(1);
        public void map(Object key, Text value, Context context) throws IOException, InterruptedException {
            String[] columns = value.toString().split("\t");
            if (columns.length > 1 && !columns[0].equals("marketplace")) {
                context.write(new Text(columns[1]), one); // Key: Customer ID
            }
        }
    }

    public static class TopReviewerReducer extends Reducer<Text, IntWritable, Text, IntWritable> {
        private String bestReviewer = "";
        private int maxReviews = 0;
        public void reduce(Text key, Iterable<IntWritable> values, Context context) {
            int sum = 0;
            for (IntWritable val : values) { sum += val.get(); }
            if (sum > maxReviews) {
                maxReviews = sum;
                bestReviewer = key.toString();
            }
        }
        protected void cleanup(Context context) throws IOException, InterruptedException {
            context.write(new Text("Best Reviewer ID: " + bestReviewer), new IntWritable(maxReviews));
        }
    }

    public static void main(String[] args) throws Exception {
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Top Reviewer");
        job.setJarByClass(TopReviewer.class);
        job.setMapperClass(ReviewerMapper.class);
        job.setReducerClass(TopReviewerReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}

```

Appendix B. Apache Pig Script

```
shadman@AI02SAM:~$ nano amazon_pig_analysis.pig
/* 1. Safely remove old output folders */
rmf /amazon/pig_task1;
rmf /amazon/pig_task2;
rmf /amazon/pig_task3;
rmf /amazon/pig_task4;
rmf /amazon/pig_task5;

/* 2. Load and Filter the raw data */
raw_data = LOAD '/amazon/amazon_reviews_us_Digital_Video_Games_v1_00.tsv' USING PigStorage('\t') AS
(marketplace:chararray, customer_id:chararray, review_id:chararray, product_id:chararray,
product_parent:chararray, product_title:chararray, product_category:chararray, star_rating:float,
helpful_votes:int, total_votes:int, vine:chararray, verified_purchase:chararray, review_headline:chararray,
review_body:chararray, review_date:chararray);

-- Filter out the header row and any corrupted/null data
clean_data = FILTER raw_data BY (star_rating IS NOT NULL) AND (total_votes IS NOT NULL) AND (marketplace !=
'marketplace');

/* ----- */
/* TASK 1: Top 5 Game Franchises (Product Parents) by Review Count */
/* ----- */
grp_parent = GROUP clean_data BY product_parent;
parent_counts = FOREACH grp_parent GENERATE group AS product_parent, COUNT(clean_data) AS total;
grp_all_1 = GROUP parent_counts ALL;
top_5_parent = FOREACH grp_all_1 GENERATE FLATTEN(TOP(5, 1, parent_counts));
STORE top_5_parent INTO '/amazon/pig_task1' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 2: Top 10 Rated Individual Reviews */
/* ----- */
-- Sort by star_rating (column index 7)
grp_all_2 = GROUP clean_data ALL;
top_10_rated = FOREACH grp_all_2 GENERATE FLATTEN(TOP(10, 7, clean_data));
STORE top_10_rated INTO '/amazon/pig_task2' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 3: Top 10 Rated Game Franchises */
/* ----- */
avg_rate = FOREACH grp_parent GENERATE group AS product_parent, AVG(clean_data.star_rating) AS avg_rating;
grp_all_3 = GROUP avg_rate ALL;
top_10_avg_rate = FOREACH grp_all_3 GENERATE FLATTEN(TOP(10, 1, avg_rate));
STORE top_10_avg_rate INTO '/amazon/pig_task3' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 4: Top 10 Most Helpful Individual Reviews */
/* ----- */
-- Sort by helpful_votes (column index 8)
grp_all_4 = GROUP clean_data ALL;
top_10_helpful = FOREACH grp_all_4 GENERATE FLATTEN(TOP(10, 8, clean_data));
STORE top_10_helpful INTO '/amazon/pig_task4' USING PigStorage('\t');
EXEC;

/* ----- */
/* TASK 5: Top 10 Most Interacted Game Franchises (Total Votes) */
/* ----- */
sum_votes = FOREACH grp_parent GENERATE group AS product_parent, SUM(clean_data.total_votes) AS
total_franchise_votes;
grp_all_5 = GROUP sum_votes ALL;
top_10_sum_votes = FOREACH grp_all_5 GENERATE FLATTEN(TOP(10, 1, sum_votes));
STORE top_10_sum_votes INTO '/amazon/pig_task5' USING PigStorage('\t');
EXEC;
```

Appendix C. Selected Execution Evidence

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_stats/part-r-00000 | head -n 20
B000AQ7K4I      MaxRate: 5.0      HelpfulVotes: 7 TotalVotes: 7
B001AU6TQ8      MaxRate: 5.0      HelpfulVotes: 38      TotalVotes: 44
B001AUEITS      MaxRate: 5.0      HelpfulVotes: 45      TotalVotes: 51
B001E3RHA8      MaxRate: 5.0      HelpfulVotes: 0 TotalVotes: 0
B001IVC66G      MaxRate: 5.0      HelpfulVotes: 0 TotalVotes: 0
B001JP5H26      MaxRate: 5.0      HelpfulVotes: 74      TotalVotes: 108
B001K7HUQY      MaxRate: 5.0      HelpfulVotes: 3 TotalVotes: 6
B001K7HUR8      MaxRate: 5.0      HelpfulVotes: 69      TotalVotes: 121
B001K7HUSW      MaxRate: 5.0      HelpfulVotes: 0 TotalVotes: 0
B001K7HUT6      MaxRate: 5.0      HelpfulVotes: 1 TotalVotes: 1
B001K7HUTQ      MaxRate: 5.0      HelpfulVotes: 0 TotalVotes: 0
B001K7HUUK      MaxRate: 5.0      HelpfulVotes: 0 TotalVotes: 1
B001K7HUUU      MaxRate: 5.0      HelpfulVotes: 17      TotalVotes: 19
B001K7HUV4      MaxRate: 4.0      HelpfulVotes: 0 TotalVotes: 0
B001K7HUVY      MaxRate: 5.0      HelpfulVotes: 0 TotalVotes: 0
B001K7HUWI      MaxRate: 4.0      HelpfulVotes: 2 TotalVotes: 2
B001K7HUWS      MaxRate: 5.0      HelpfulVotes: 16      TotalVotes: 17
B001K7HUX2      MaxRate: 5.0      HelpfulVotes: 11      TotalVotes: 15
B001K7HUXC      MaxRate: 3.0      HelpfulVotes: 1 TotalVotes: 1
B001K7HUXM      MaxRate: 4.0      HelpfulVotes: 0 TotalVotes: 0
cat: Unable to write to output stream.
```

C.1 GameStats output reproduced from the supplied raw report.

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_avg/part-r-00000 | head -n 20
B000AQ7K4I      Moving Averages: 4.0,4.5,
B001AU6TQ8      Moving Averages: 3.0,2.0,1.6666666,1.5,1.4,2.0,1.8571428,1.75,2.1111112,2.0,1.9090909,1.8333334,1.7692307,
B001AUEITS      Moving Averages: 4.0,2.5,3.0,2.5,2.2,2.0,2.2857144,2.625,2.4444444,2.3,2.4545455,2.4166667,2.4615386,2.357143,2.4666667,2.375,2.470588
2,
B001E3RHA8      Moving Averages: 5.0,
B001IVC66G      Moving Averages: 5.0,
B001JP5H26      Moving Averages: 4.0,2.5,3.3333333,2.75,2.4,2.8333333,2.5714285,2.75,2.5555556,2.4,2.2727273,2.4166667,2.5384614,2.4285715,2.3333333,2
.5,2.5882354,2.5555556,2.4736843,2.4,2.3333333,2.4545455,2.5652175,
B001K7HUQY      Moving Averages: 5.0,5.0,3.6666667,
B001K7HUR8      Moving Averages: 4.0,4.5,4.6666665,4.75,4.8,4.6666665,4.571429,4.125,4.0,4.0,4.090909,4.0833335,4.0,4.0,4.0,4.0625,4.0,4.0,3.8947368,3
.85,3.9047618,3.8636363,3.9130435,3.9583333,4.0,3.9230769,3.9629629,3.9285715,3.9655173,4.0,4.032258,4.0625,3.969697,3.9705882,3.942857,3.9444444,3.97
29729,4.0,3.9230769,3.9,3.902439,3.9047618,3.8372092,3.8636363,3.8888888,3.826087,3.8510637,3.875,3.8979592,3.84,3.8431373,3.8269231,
B001K7HUSW      Moving Averages: 5.0,4.5,3.6666667,
B001K7HUT6      Moving Averages: 5.0,
B001K7HUTQ      Moving Averages: 1.0,3.0,
B001K7HUUK      Moving Averages: 5.0,4.5,3.3333333,
B001K7HUUU      Moving Averages: 5.0,5.0,4.6666665,4.5,4.6,4.6666665,4.571429,4.5,4.2222223,4.3,4.0,4.0833335,4.076923,4.142857,
B001K7HUV4      Moving Averages: 2.0,3.0,
B001K7HUVY      Moving Averages: 4.0,2.5,3.3333333,
B001K7HUWI      Moving Averages: 4.0,
B001K7HUWS      Moving Averages: 2.0,3.0,3.6666667,3.75,3.8,4.0,4.142857,4.25,
B001K7HUX2      Moving Averages: 4.0,4.5,3.3333333,2.75,3.2,2.8333333,3.142857,3.0,3.2222223,3.3,3.2727273,3.4166667,3.5384614,3.642857,
B001K7HUXC      Moving Averages: 3.0,
B001K7HUXM      Moving Averages: 4.0,
cat: Unable to write to output stream.
```

C.2 MovingAverage output reproduced from the supplied raw report.

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/output_best/part-r-00000
Best Reviewer ID: 21925455      95
```

C.3 TopReviewer output reproduced from the supplied raw report.

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/pig_task1/part-r-00000
402222185      3244
395682204      3503
341969535      3438
384246568      13642
232803743      7307
```

C.4 Pig Task 1 output reproduced from the supplied raw report.

US	31525534	R2GVSDHW513SS1	B002LIT9EC	695277014	Build-a-lot 4: Power Source [Download]	Digital_Video_Games	5.0	0	0	N	Y	Probably the b
US	133437	R1WFOQ3N9B065I	B00F4CEHNK	341969535	Xbox Live Gift Card	Digital_Video_Games	5.0	0	0	N	Y	Five Stars
US	45331959	R3V3N1JLZM0E552	B001QCWQCQ	704016361	Mushroom Age [Download]	Digital_Video_Games	5.0	9	10	N	N	One of the best hide-n-s
US	49511256	RUHZQA7YBHM	B001KC01WO	955209542	Build-a-lot [Download]	Digital_Video_Games	5.0	1	4	N	N	Awesome Game!!! This
US	6057518	R44QKV6FE5CJ2	B004RMK4BC	384246568	Playstation Network Card	Digital_Video_Games	5.0	0	0	N	Y	Five Stars Super
US	30331331	R3U5406SKFSPJH	B001KC0070	137332584	Dangerous High School Girls in Trouble! [Download]	Digital_Video_Games	5.0	4	5	N	N	N
US	27224973	R7K2JVA47V6KQ	B001VTQIDY	225309182	2009-02-03							
US	52062256	R3QZBHD7V6V1RK	B001KC00HU	631385531	Jewel Quest 2 [Download]	Digital_Video_Games	5.0	27	45	N	N	Microsoft points
US	50240227	R2ZNOZCXGUEPV8	B001LSTIUG	536943092	The Scruffs [Download]	Digital_Video_Games	5.0	1	2	N	N	platform-specific
US	37218186	R14UC11W5PMD3U	B001K7HV1S	706196778	Battleship Chess [Download]	Digital_Video_Games	5.0	4	4	N	N	Hey, Good Family game

C.5 Pig Task 2 output reproduced from the supplied raw report.

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/pig_task3/part-r-00000
12720703      5.0
400079      5.0
993689839      5.0
993119997      5.0
4567536      5.0
996136365      5.0
998464095      5.0
997289579      5.0
994878730      5.0
996288895      5.0
```

C.6 Pig Task 3 output reproduced from the supplied raw report.

US	52990130	R1Q4ELO9TG7JCW	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	1.0	685	735	N	N	I wish I could actually pl
US	12708262	R1DIKEVOTQNCB	B002VBWIP6	232803743	Xbox Live Subscription	Digital_Video_Games	5.0	993	1077	N	Y	Better than buying one i
US	34790787	R4J30P1CV021Y	B004RMK4BC	384246568	Playstation Network Card	Digital_Video_Games	5.0	882	959	N	Y	Can Be Purchased Outsid
US	44692602	R281YKQ2AM70KM	B004RMK4P8	384246568	Playstation Network Card	Digital_Video_Games	5.0	2384	2541	N	N	Why buy the \$50 when
US	29805145	R2Q8KUKOZ9LZX	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	1.0	1100	1151	N	N	One of the saddest mor
US	13560072	R12CY204A3M00	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	1.0	1139	1212	N	N	Simcity, an overpriced 2i
US	26458957	R3TD4ZVF0G6OON	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	1.0	1823	1943	N	N	Now Origin has become
US	37248460	RH1PGSV6QJME	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	1.0	3789	3948	N	N	Singleplayer DRM does
US	26834793	R3MHPUOS7SBOHK	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	1.0	5068	5251	N	Y	Hey everyone!
US	50432755	R2MC4X10C5KZFO	B007VTVRFA	402222185	SimCity - Limited Edition	Digital_Video_Games	5.0	2509	2974	N	Y	Got me off my video ga

C.7 Pig Task 4 output reproduced from the supplied raw report.

```
shadman@AI02SAM:~$ hdfs dfs -cat /amazon/pig_task5/part-r-00000
483903773      3519
289370895      3584
386077723      3888
240448759      4234
28899677       3613
954148166      4107
232803743      4792
395682204      4558
402222185      46679
384246568      8515
```

C.8 Pig Task 5 output reproduced from the supplied raw report.